

Retail Customer Analytics: Consumer Behaviour, Segmentation & Sales Insights

Executive Summary:

This project was conducted to analyze customer shopping behavior and identify factors influencing purchasing decisions, customer loyalty, and product preferences. Using Python for data preparation, PostgreSQL for business analysis, and Power BI for visualization, the project transformed raw retail transaction data into actionable business insights that can support marketing optimization, customer engagement, and revenue growth.

Business Problem:

A retail company wanted to understand customer purchasing behavior across demographics, product categories, discounts, subscriptions, and shopping channels. The objective was to identify customer trends, improve engagement strategies, optimize product offerings, and increase long-term customer loyalty.

Project Objectives

1. Analyze customer purchase behavior.
2. Identify high-value customer segments.
3. Evaluate impact of discounts and subscriptions.
4. Discover top-performing products.
5. Measure revenue contribution across demographics.
6. Develop an interactive business dashboard.

Dataset Overview

Metric	Value
Records	3900
Features	18
Domain	Retail / E-commerce
Tools	Python, PostgreSQL, Power BI

Features included:

- Age
- Gender
- Location
- Subscription Status
- Product Category
- Purchase Amount
- Review Rating
- Discounts
- Shipping Type
- Purchase Frequency

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Python Data Preparation

Data Loading:

[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Visa
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	MasterCard
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Visa
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	MasterCard
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	MasterCard

Data Exploration:

Purpose

- Understand data types.
- Identify missing values.
- Review distributions and categorical

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

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Missing Value Treatment:

Method

Filled missing ratings using the median rating of each product category.

```
: df.isnull().sum()
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

Column Standardization:

Purpose

- Improve readability.
- Ensure SQL compatibility.
- Follow naming conventions.

```
: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns = {'purchase_amount_(usd)': 'purchase_amount'})

: df.columns

: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='object')
```

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Feature Engineering – Age Group:

[11]:	age	age_group
0	55	Middle=Aged
1	19	Young Adult
2	50	Middle=Aged
3	21	Young Adult
4	45	Middle=Aged
5	46	Middle=Aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle=Aged

Feature Engineering – Purchase Frequency:

Purpose

Convert categorical purchase behaviour into measurable numerical values.

[13]:	purchase_frequency_days	frequency_of_purchases
0	NaN	Fortnightly
1	NaN	Fortnightly
2	NaN	Weekly
3	NaN	Weekly
4	365.0	Annually
5	NaN	Weekly
6	90.0	Quarterly
7	NaN	Weekly
8	365.0	Annually
9	90.0	Quarterly

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Redundancy Check:

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
print(df.columns)
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'promo_code_used', 'previous_purchases',  
      'payment_method', 'frequency_of_purchases', 'age_group',  
      'purchase_frequency_days'],  
      dtype='object')
```

Database Integration:

Purpose

Store cleaned data for SQL analysis.

```
[19]: from sqlalchemy import create_engine
```

```
[20]: from sqlalchemy import create_engine
```

```
print(create_engine)
```

```
<function create_engine at 0x000001A69E9482C0>
```

```
[21]: from sqlalchemy import create_engine
```

```
username = "postgres"
```

```
password = "1234"
```

```
host = "localhost"
```

```
port = "5432"
```

```
database = "customer_behavior"
```

```
engine = create_engine(
```

```
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
```

```
)
```

```
table_name = "customer" # choose any table name
```

```
df.to_sql(table_name, engine, if_exists="replace", index=False)
```

```
print("Connection object created successfully!")
```

```
Connection object created successfully!
```

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SQL Data Analysis (PostgreSQL)

Objective

After cleaning and transforming the dataset in Python, the processed data was loaded into PostgreSQL. SQL queries were used to analyse customer behaviours, purchasing patterns, product performance, customer loyalty, and revenue trends.

Database Design

Table Used

customer

Key Columns:

- customer_id
- gender
- age
- age_group
- item_purchased
- category
- purchase_amount
- review_rating
- subscription_status
- discount_applied
- shipping_type
- previous_purchases

Table Output

Data Output													
Showing rows: 1 to 10 of 1													
id	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	previous_purchases
1	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes
2	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes
3	Male	Jeans	Clothing	73	Massachu...	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes
4	Male	Sandals	Footwear	90	Rhode Isla...	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes
5	Male	Blouse	Clothing	49	Oregon	M	Turquois...	Spring	2.7	Yes	Free Shipping	Yes	Yes
6	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes
7	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes
8	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes
9	Male	Coat	Outerwear	97	West Virgi...	L	Silver	Summer	2.6	Yes	Express	Yes	Yes
10	Male	Handbag	Accessori...	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes

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Revenue by Gender

- **Objective**

Determine whether male or female customers contribute more revenue.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Business Insight

This analysis identifies the highest revenue-generating customer group and supports targeted marketing strategies.

High-Spending Discount Users

- **Objective**

Identify customers who used discounts but still spent above the average purchase amount.

Data Output	Messages	Notifications
	 	    
	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
Total rows: 839		
Query complete 00:00:00.409		

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Business Insight

These customers represent valuable segments that remain profitable even when promotional discounts are offered.

Top Rated Products

- **Objective**

Identify products receiving the highest customer ratings.

Data Output

Messages

Notifications

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SQL

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Business Insight

Highly rated products can be prioritized in marketing campaigns and inventory planning.

Shipping Type Performance

- **Objective**

Compare spending behaviour across shipping methods.

Data Output

Messages

Notifications

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

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Business Insight

Understanding shipping preferences helps optimize logistics and customer experience.

Subscriber vs Non-Subscriber Analysis

▪ Objective

Determine whether subscribers generate more revenue.

Data Output					Messages	Notifications
	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric		
1	No	2847	59.87	170436.00		
2	Yes	1053	59.49	62645.00		

Business Insight

Results help evaluate the effectiveness of subscription programs and customer retention initiatives.

Discount-Dependent Products

▪ Objective

Identify products most frequently purchased using discounts.

Data Output			Messages	Notifications
	item_purchased text	discount_rate numeric		
1	Hat	50.00		
2	Sneakers	49.66		
3	Coat	49.07		
4	Sweater	48.17		
5	Pants	47.37		

Business Insight

These products may require promotional support to drive sales.

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Customer Segmentation

▪ Objective

Classify customers based on purchase history.

Data Output		Messages	Notifications
			
			
	customer_segment text	Number of Customers bigint	
1	Loyal	3116	
2	New	83	
3	Returning	701	

Business Insight

Customer segmentation helps create personalized marketing and loyalty strategies.

Top Products within Categories

▪ Objective

Identify the most popular products in each category.

Data Output

Messages

Notifications

Business Insight

Top-selling products can be prioritized for promotions and inventory management.

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Repeat Buyers and Subscription Status

▪ Objective

Examine whether repeat buyers are more likely to subscribe.

Data Output		Messages	Notifications
	subscription_status text		repeat_buyers bigint
1	No		2518
2	Yes		958

Business Insight

This analysis measures the relationship between customer loyalty and subscription adoption.

Revenue Contribution by Age Group

▪ Objective

Determine which age segments generate the highest revenue.

Data Output		Messages	Notifications
	age_group text		total_revenue numeric
1	Young Adult		62143
2	Middle=Ag...		59197
3	Adult		55978
4	Senior		55763

Business Insight

High-revenue age groups can be targeted with personalized campaigns and product recommendations.

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Dashboard Development & Visualization (Power BI)

Objective

The Power BI dashboard was developed to transform analytical results into interactive visual insights. It enables stakeholders to monitor customer behavior, revenue trends, subscription performance, product categories, and demographic patterns for better decision-making.

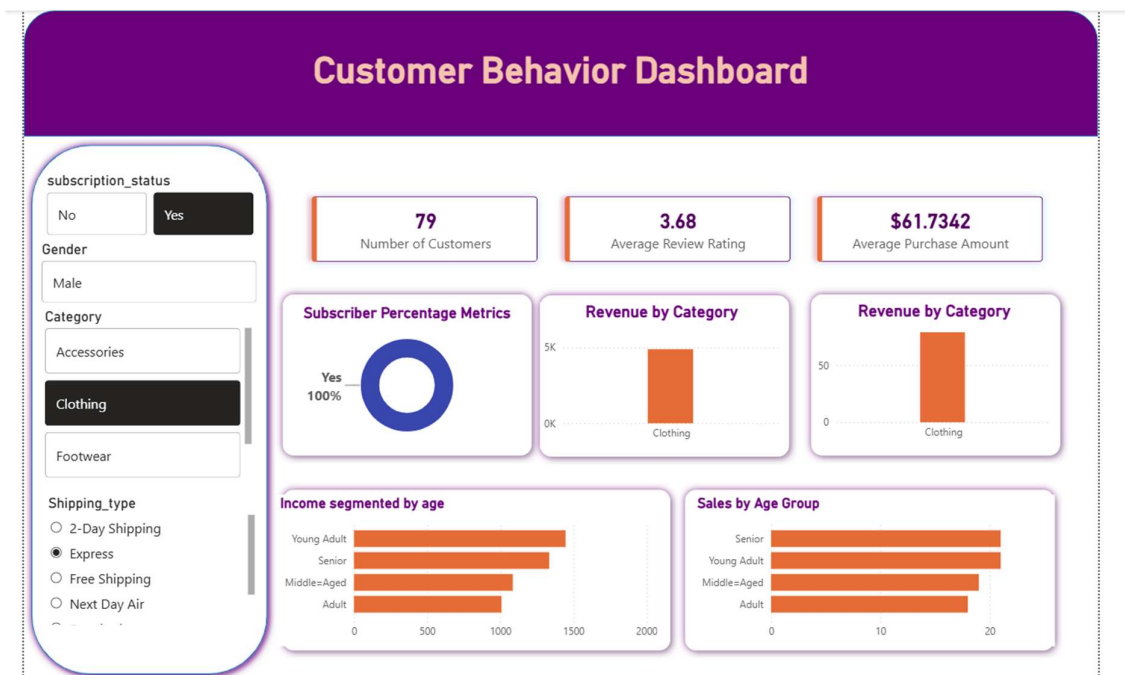
Dashboard Features

The dashboard includes dynamic slicers allowing users to filter data by:

- Subscription Status
- Gender
- Product Category
- Shipping Type

Business Benefit

Allows management to drill down into specific customer segments and analyse purchasing behaviours.



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Business Recommendations

Increase Subscription Adoption

Offer exclusive benefits and loyalty rewards to attract more subscribers.

Focus on High-Revenue Categories

Invest marketing resources in top-performing categories.

Personalized Marketing

Target customers based on age groups, purchase behavior, and spending patterns.

Customer Retention Programs

Reward repeat buyers through loyalty points and membership benefits.

Product Promotion Strategy

Feature highly rated products in campaigns and promotional activities.